

Ethan Karvelsson

Mechanical Engineering Student

CONTACT INFORMATION	SUMMARY
Bozeman, Montana (346) 372-6510 x92f224@student.montana.edu LinkedIn.com/in/ethan-karvelsson-008b13367 Ethan Karvelsson Engineering Portfolio	I am an aspiring mechanical engineering student at Montana State University with a focus on learning about turbomachinery and propulsion applications in the aerospace industry. I am a hands-on learner applying the knowledge I learn in my classes, labs and independent research to plan and build projects and experiments to solidify my understanding of the theory involved in physical systems. My engineering design philosophy involves researching operational theory, determining design constraints, creating initial prototypes, then extensive iteration from the prototypes to create unique solutions for design and physical problems that arise in the process.
CORE SKILLS	PROFESSIONAL EXPERIENCE
• Extensive SolidWorks Experience • Python Programming • MATLAB Programming • Ansys Fluent Simulations • Sheet Metal Fabrication • Manual Lathe Experience • Manual Mill Experience • TIG Welding – Stainless Steel/Steel • Electronic Circuit Design and Construction. • Soldering & Electrical Wiring Construction. • Hydraulic Systems Design • Fuel Systems Design	President of the American Society of Experimental Aircraft Club [ASEA] @ Montana State University <i>05/2026 – Present</i> • Establishing leadership and direction for the ASEA club, AKA “the rocket club”. • Running admin and operations. • Planning events and meeting schedules. Lead Ramp Agent – WFS Bozeman International Airport <i>05/2025 – Present</i> • Providing core support to the ground operations for Delta Airlines, JetBlue, Allegiant, Sun Country and Charter Flights. • Leading ground service operations for these airlines at the BZN international airport. • Providing leadership to fellow employees and most importantly ensuring the safety of those working under my leadership. • Conducting aircraft pushback operations. • Loading aircraft luggage according to load plans created by supervisors running ALIS (Automated Load Instruction Service). • Verifying position and weight of loaded luggage is within safe operational limits of aircraft. • Sending AWABS (Automated Weight and Balance System) data to flight crew before departure. • Inspecting aircraft for damage upon arrival and before departure. • Intensive time management and operations planning involved turning flights in and out of gates on time. • Balancing safety with time management/performance. • Adaptability to the constantly changing operating conditions on the ramp.

Volunteering - Lonestar Flight Museum

10/2024 – 12/2024

- Learned about volunteering operations that keeps one of the largest fleets of actively flying World War 2 era aircraft in the world operational.

CERTIFICATIONS	EDUCATION
• Ramp ALA (Aircraft Lead Agent) Certified through WFS and Delta airlines. • Pushback Operator Certified through WFS and Delta Airlines. • De-Ice Operations Certified through WFS and Delta Airlines. • Pursuing L1 – High Powered Rocket Launch Certification on October 17th, 2026. (Weather Dependent)	Junior studying for B.S. Mechanical Engineering, pursuing a minor in Mechatronics — Montana State University (2025 – Present)
Citizenship	PROJECT HIGHLIGHTS
• Full US Citizenship	• Design and construction of a twin-spool radial turbojet engine called Typheonix. Created from two automotive turbochargers arranged in a compound or series configuration. • CFD analysis of the combustion chamber for the Typheonix Engine. • 3D Surface plots in MATLab, representing a function yielding compressor drive power from multiple input variables, created using data from compressor maps from the turbocharger manufacturer - BorgWarner, for the Typheonix Engine. • Detailed CAD modeling of a turbocharger in SolidWorks. • Design and construction of a high-power capacity three channel analog electric control box. • Design and construction of a high-voltage high-power electric power supply and capacitor bank. • Testing of small-scale combustion chamber prototype using compressed air and methanol fuel. • High-speed video study of shockwaves from a controlled gas explosion. • Chemical synthesis of a copper-based oxidizing compound that deflagrates (burns rapidly) with a blue flame.